Early-cycle intensity correction and calibrated intensity ensembles for physical-model TC guidance

Project write-up · status as of 29 October 2026 · North Atlantic and East Pacific

Summary. This project started from a simple question raised by AIFS-TC (Allen et al. 2026): if a cheap learned correction can lift an AI weather model's tropical-cyclone intensity forecasts to the operational frontier, can the same idea add value on top of the *physical* models (GFS/GEFS, IFS/ECMWF ensemble, HAFS) that forecasters already use, using only public data and only information that is available in time for the forecast? The work has produced a three-part system. A **consensus correction** learns, with gradient-boosted trees, how far the average of NHC's early guidance typically is from the eventual best track, given the storm's current state and the disagreement between the guidance. An optional **gridded-feature** branch adds information extracted from the GEFS and IFS ensemble fields themselves. Finally, a **calibrated ensemble** layer turns each raw physical ensemble (31 GEFS or 51 ECMWF members), whose intensities are far too weak and far too similar, into a set of smooth, realistic intensity scenarios that are centred on the corrected consensus and spread according to how uncertain such forecasts have historically been.

Everything is tested on the combined 2025 and 2026 seasons, held out from training. Testing on 2026, the first season run as it would be in real time, showed that NHC's public real-time files carry no ECMWF aids. The consensus correction is therefore trained without ECMWF inputs, which also made it more accurate in 2025. In 2025 it matches NHC's corrected consensus HCCA, and over 2018–2024 it beats HCCA and the official forecast on rapid intensification. In 2026, however, it falls about 2 kt behind HCCA (which uses ECMWF guidance internally) and is several knots too weak. Gridded fields from our own IFS-ENS extraction, the only real-time route to ECMWF information, now reduce the consensus error by 1–1.5 kt; GEFS fields add nothing, even with five seasons of training data.

The result is two real-time ensemble products. **GECO** (GEFS Ensemble, Consensus-Oriented) reshapes NHC's public GEFS member tracks around HCCA at every cycle; **IECO** does the same for our IFS-ENS members at 12 UTC. Neither needs any gridded data in its centre or spread. Because GECO needs only NHC's files, it was tested on nine seasons (2018–2026, 336 storms): it beats a probabilistic version of HCCA in every season, by 0.36 kt of CRPS on average, with about three quarters of outcomes inside its 10–90% range. Over 2025–26 it beats FNV3′s members by 0.78 kt (significant) and is nearly unbiased. IECO, on the 12 UTC cycles, is 0.65 kt better than FNV3′s members (not significant) and a little behind GECO. As for every system, rapidly intensifying storms remain the main weakness.

1 Motivation and design principles

AIFS-TC corrected the very weak intensities of ECMWF's AIFS-Single AI model with a gradient-boosted tree and a small convolutional network trained on nine seasons of hindcasts, cutting the model's error by about a factor of three and bringing it level with FNV3 and the NHC official forecast. The physical models used operationally are much less biased than AIFS at the storm scale, but their ensembles, which are run at coarse resolution, still produce intensities that are far too weak and members that disagree far too little. The question here is therefore not only whether a learned correction helps deterministic intensity forecasts from these models, but also whether it can turn their ensembles into usable probabilistic intensity guidance.

Four principles shaped the design. First, everything is built for the **early cycle**: an input may only be used if it is available within about three hours of the synoptic time, exactly as for NHC's own consensus aids and the official forecast. This makes the comparison with OFCL fair (the AIFS-TC paper compared a late model) and means the system could run operationally. Second, only **public data** are used: NHC's forecast and best-track files, IBTrACS, NCAR's archive of real-time SHIPS output, model archives from dynamical.org, ECMWF

open data and Google DeepMind's Weather Lab archive. Third, the work is **staged**, starting with the cheapest version that could possibly work (track files only) and adding gridded data only to measure what it contributes. Fourth, all comparisons follow NHC's verification conventions and use **homogeneous** samples with storm-block bootstrap confidence intervals, so that differences between systems are measured on identical cases and judged against the storm-to-storm variability of a single season (details in Appendix A).

2 System architecture

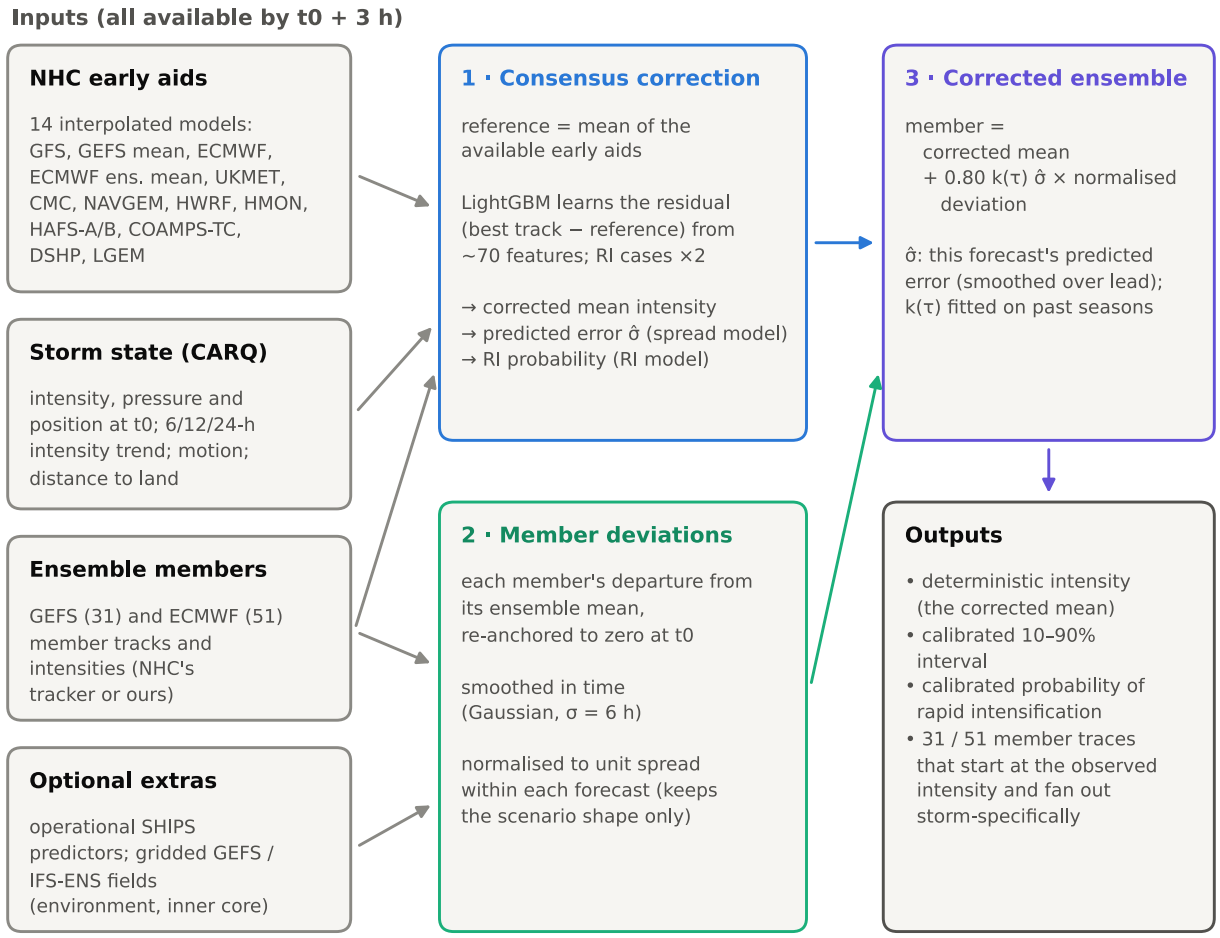


Figure 1: Overview of the system. All inputs are available within three hours of the synoptic time t_0. The consensus correction (1) produces a corrected mean intensity, a predicted error and a probability of rapid intensification; the member deviations (2) carry each physical ensemble's scenario structure; the corrected ensemble (3) combines the two.

Figure 1 shows how the pieces fit together. At each synoptic time t_0 the system gathers what NHC's forecasters would have in hand: the operational analysis of the storm (CARQ), the interpolated ("early") versions of the dynamical and statistical-dynamical models that NHC publishes in real time, and the member tracks of the GEFS ensemble from the most recent run that is old enough to have arrived. ECMWF guidance enters only through our own extraction from ECMWF open data, because NHC's real-time files carry no ECMWF aids. The **consensus correction** forms a simple reference forecast, the mean of whichever early aids are available, and uses a gradient-boosted tree model to predict how far the best track will be from that reference at each lead time. Its output is a corrected mean intensity, plus a 10–90% interval derived from quantile models and widened by conformal calibration so that it holds its nominal coverage.

The **ensemble layer** starts from a raw physical ensemble. Each member's departure from its own ensemble mean is treated as a scenario: it is re-anchored so that all members start together at the observed intensity,

lightly smoothed in time so that it evolves like a real storm, and normalised so that only its shape (which members intensify faster or slower, and when) is kept. The size of the spread comes instead from a **spread model**, trained on the consensus correction's past errors, that predicts how wrong this particular forecast is likely to be. The **corrected ensemble** is a consensus centre plus the normalised scenarios scaled by that predicted error. For both products, GECO (GEFS) and IECO (IFS-ENS), the centre is NHC's corrected consensus HCCA, which is in the public real-time files. A companion **RI model** gives a calibrated probability of rapid intensification. A separate **gridded-feature** branch (section 5) can feed additional predictors from the ensemble fields into the consensus correction.

All models are trained season-wise. The consensus correction uses 2018–2024 with leave-one-season-out cross-validation and is tested on 2025 and 2026 combined (about 60 named storms). The 2025 season is also the test year of the AIFS-TC paper, which makes those results directly comparable. The 2026 season, verified against NHC's working best track up to 24 October, is the first season run exactly as in real time, from NHC's public real-time files. The gridded models are cross-validated by season over what the archive allows: IFS-ENS 2024–2026 and GEFS 2020–2026.

3 Inputs and variables

3.1 Storm state and early guidance

The storm state comes from NHC's operational analysis (CARQ) at t_0: maximum wind and minimum pressure, position, the intensity change over the previous 6, 12 and 24 hours, the storm's motion over the previous 12 hours and the distance to land. The guidance consists of the interpolated aids listed in Table 1. Interpolated aids are the previous cycle's late model runs shifted in time and adjusted to the current intensity; following NHC's verification practice, the 6-hour and 12-hour versions of each aid are merged into one. For each aid the model sees its intensity at the lead time in question and its change over the preceding 12 hours; missing aids are simply left missing. The ECMWF aids are in NHC's archive files but absent from its real-time files during the season, so they are not used. A version trained with them was 0.65 kt worse over 2025–26 (significant), because it learned to lean on inputs that are missing in real time; even in 2025, when the archive has them, it was 0.4 kt worse (`docs/m15_realtime.md`).

Aid	Model	Type
AVNI, AEMI	GFS deterministic; GEFS ensemble mean	global, physical
EGRI, CMCI, NVGI	UK Met Office; Canadian; US Navy	global, physical
HWFI, HMNI, HFAI, HFBI, CTCI	HWRF; HMON; HAFS-A; HAFS-B; COAMPS-TC	regional hurricane models
DSHP, LGEM	SHIPS with decay over land; Logistic Growth Equation Model	statistical-dynamical

Table 1: The twelve early (interpolated) aids used by the consensus correction. ECMWF's EMXI and EMNI are not used, because they are absent from NHC's real-time files.

3.2 Ensemble members

The GEFS (30 perturbed members) and ECMWF (50 perturbed members) ensembles enter in two ways. NHC's tracker output for the GEFS members is in NHC's real-time files for every cycle; to respect the early-cycle rule we use the newest run that is at least 6 hours old, with lead times shifted accordingly. NHC's ECMWF member tracks exist only in the archive, so they are used just for a 2025 comparison of ensembles, not as inputs. For the gridded pilot and the corrected ensembles we also ran our own tracker on the full ensembles (31 GEFS and 51 IFS-ENS members including the control), so that the member intensities, structure and environment are all derived consistently from the model fields. In the consensus correction the members are summarised by their mean, spread, 10th and 90th percentiles, the fraction of members that intensify by at least 30 kt in 24 hours, and their offset from the observed intensity at t_0.

3.3 Optional predictors: SHIPS and gridded fields

The real-time SHIPS output archived by NCAR provides environmental predictors computed along the previous official forecast track: vertical wind shear and its direction, sea-surface temperature and the corresponding potential intensity, 200 hPa temperature, mid-level relative humidity, 850 hPa vorticity, 200 hPa divergence, ocean heat content, the model vortex strength and satellite infrared predictors. The gridded branch reads surface fields for every ensemble member around the storm (sea-level pressure, 10 m wind and gusts, 2 m temperature and humidity, low- and mid-level heights and temperatures for IFS-ENS, precipitable water for GEFS) and pressure-level fields from one member for the environment (winds at 200, 500 and 850 hPa, temperatures at 200 and 500 hPa, humidity at 500–850 hPa, vorticity, divergence, skin temperature and column water). From these we derive, for each member and lead time, the tracked centre, minimum pressure, maximum wind and gust, radius of maximum wind, the depth of the pressure minimum and inner-core-minus-surroundings contrasts (warm core, moist core), and for the ensemble as a whole SHIPS-style environmental quantities around the ensemble-mean centre (shear, steering flow, humidity, vorticity, divergence, SST and an empirical maximum potential intensity). Table 2 summarises the feature groups.

Group	Count	Examples
CARQ / storm state	10	vmax0, pmin0, lat0, lon0, dv6, dv12 …
Interpolated aids (value, 12-h change)	24	AEMI_vmax, AVNI_vmax, CMCI_vmax, CTCI_vmax, DSHP_vmax, EGRI_vmax …
GEFS members (AP01–30)	11	gefs_vmax_mean, gefs_vmax_std, gefs_vmax_p10, gefs_vmax_p90, gefs_n, gefs_mslp_mean …
ECMWF members (EE01–50)	0	
Consensus summary	8	n_aids, ref_vmax, ref_pmin, aid_vmax_std, aid_vmax_range, lat_c …
Calendar / basin / lead	3	lead, basin_ep, doy
SHIPS (variant B only)	39	ships_shear, ships_shear_adj, ships_shear_dir, ships_sst, ships_pot_int, ships_t200 …

Table 2: Feature groups of the consensus correction (variant A uses all but SHIPS; variant B adds SHIPS). The gridded branch adds roughly 30–60 further features per model.

4 Consensus correction

4.1 Formulation

For each storm, synoptic time t_0 and lead time $\tau \in \{12, 24, \ldots, 168\}$ h the model predicts the residual between the best-track intensity and the reference consensus,

$$\hat{v}(t_0+\tau) = \overline{v}_{\mathrm{ref}}(t_0,\tau) + f(\boldsymbol{x}), \quad f \text{ trained on } v_{\mathrm{obs}} - \overline{v}_{\mathrm{ref}},$$

where $\overline{v}_{\mathrm{ref}}$ is the equal-weight mean of the available early aids and $\boldsymbol{x}$ collects the features of section 3. A single LightGBM model covers all lead times (the lead is a feature), cases of rapid intensification are given twice the weight in training, as in AIFS-TC, and the hyperparameters are fixed rather than tuned (600 trees of 15 leaves, learning rate 0.03). Three additional quantile models give the 10th, 50th and 90th percentiles. Because tree models fit the training storms' variability too closely, their intervals are too narrow on new storms; a conformal adjustment, fitted per lead time on out-of-fold predictions, widens them by about 1 kt at 12 h and about 11 kt at 168 h. The same machinery predicts minimum pressure, using as reference the operational pressure at t_0 plus the GEFS ensemble's predicted pressure change, since the interpolated aids carry no pressure.

4.2 What it achieves

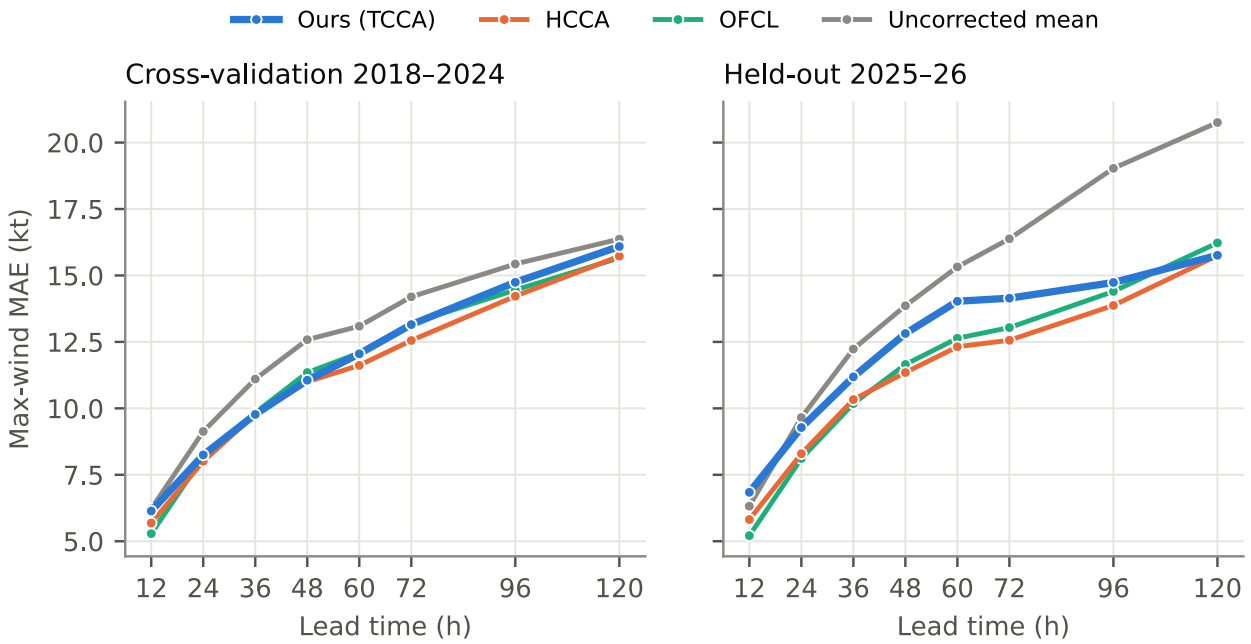


Figure 2: Maximum-wind error by lead time for the consensus correction (TCCA, without SHIPS), NHC's corrected consensus HCCA, the official forecast and the uncorrected aid mean. Left: out-of-fold 2018–2024; right: the held-out 2025–26 seasons. Identical cases for all systems.

Over the 2018–2024 cross-validation the corrected consensus is as accurate as NHC's best consensus aids and the official forecast (Figure 2): its average error differs from HCCA's by 0.3 kt [−0.09, 0.71]. The two test seasons behave differently. In 2025 the difference from HCCA is −0.23 kt [−0.96, 0.86], with almost no bias, and the correction is several knots better than HCCA and OFCL at four to five days. In 2026 it is 2.05 kt [0.26, 3.88] worse than HCCA, significantly, though level with the simple consensus IVCN (−0.05 kt [−0.74, 0.74]), and it is about 3.8 kt too weak. HCCA uses ECMWF guidance that NHC has internally but does not publish in real time, and its 2026 errors are unusually low, especially beyond two days, so part of the gap is the missing ECMWF information and part is a 2026 season in which the early guidance was weaker than usual. Over both seasons together the difference from HCCA is 1.11 kt [−0.03, 2.34]. In every sample the model removes most of the raw consensus's weakness: the uncorrected aid mean is about 6.0 kt too weak over 2025–26. The correction is weakest at 12 h, where the official forecast and HCCA benefit from subjective and objective use of the very latest observations.

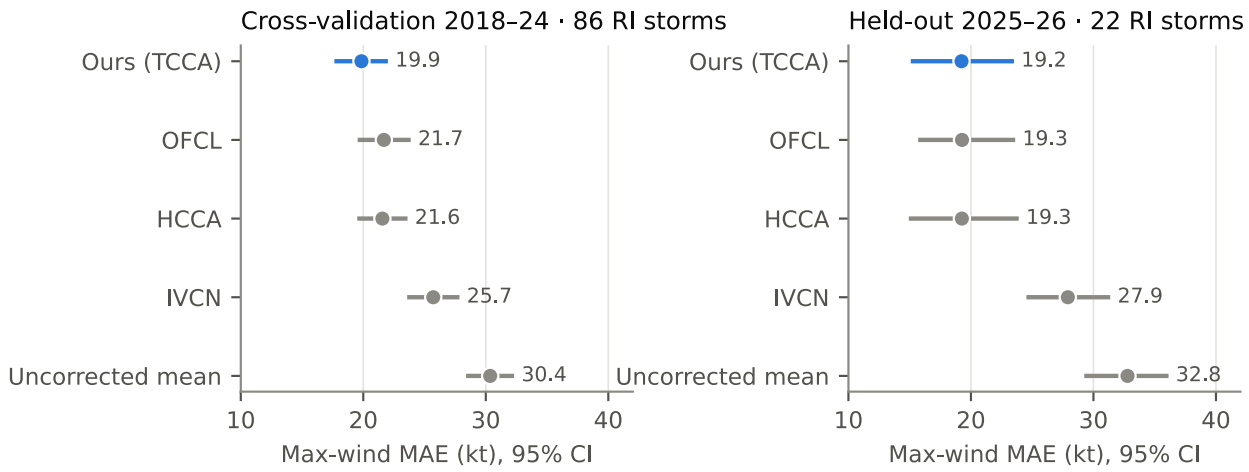


Figure 3: Error on cases of rapid intensification (an observed increase of at least 30 kt in 24 h), with 95% storm-bootstrap intervals.

The clearest gain is on rapid intensification (Figure 3). Over 2018–2024 the corrected consensus beats HCCA on these cases by −1.71 kt [−2.84, −0.63] and the official forecast by −1.83 kt [−3.24, −0.42]. In 2025–26 it is level with both (−0.03 kt [−2.85, 3.06] against HCCA). Every system, including ours, remains strongly biased low during rapid intensification, which is the main remaining weakness of intensity guidance in general. Adding the operational SHIPS predictors changes the error by less than 0.2 kt, which is not surprising because DSHP and LGEM, themselves built on SHIPS, are already inputs. Against the early version of FNV3 over 12–168 h the corrected consensus is about 0.9 kt worse on average (not significant) but less biased, and better beyond about five days.

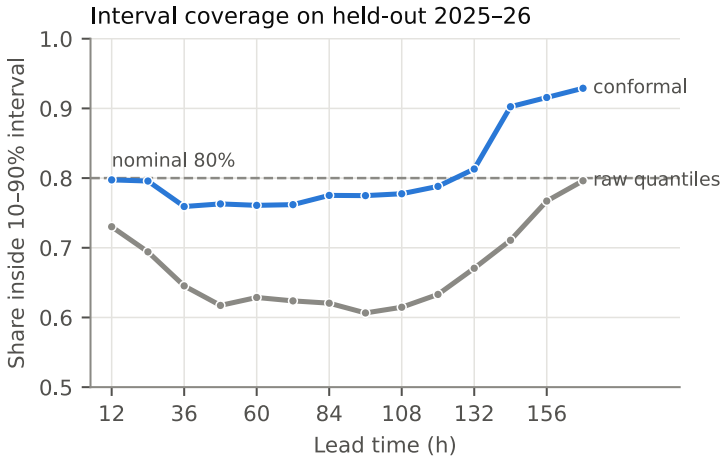


Figure 4: Share of 2025–26 outcomes inside the 10–90% interval by lead time, before and after conformal widening.

5 Gridded features from the physical ensembles

The gridded branch asks whether the ensemble fields contain information that the track files do not. We extracted every 00 UTC run of IFS-ENS from June 2024 to October 2026 and of GEFS from October 2020 to October 2026 with an active storm in either basin (347 IFS-ENS and 839 GEFS runs), tracked every member with a simple pressure-minimum tracker that agrees closely with NHC's own tracker (median position differences of 15.1 km for IFS-ENS and 32.8 km for GEFS, wind correlations of 0.98–0.99), and derived the per-member structure and ensemble-environment features described in section 3. Under the early-cycle rule the IFS-ENS 00 UTC run serves the 12 UTC forecast and the GEFS 00 UTC run the 06 UTC forecast.

The raw ensembles illustrate the problem the correction has to solve (Figure 5): their tracks are good, but their intensities are around 23 kt too weak on average and their members far too similar. Three experiments

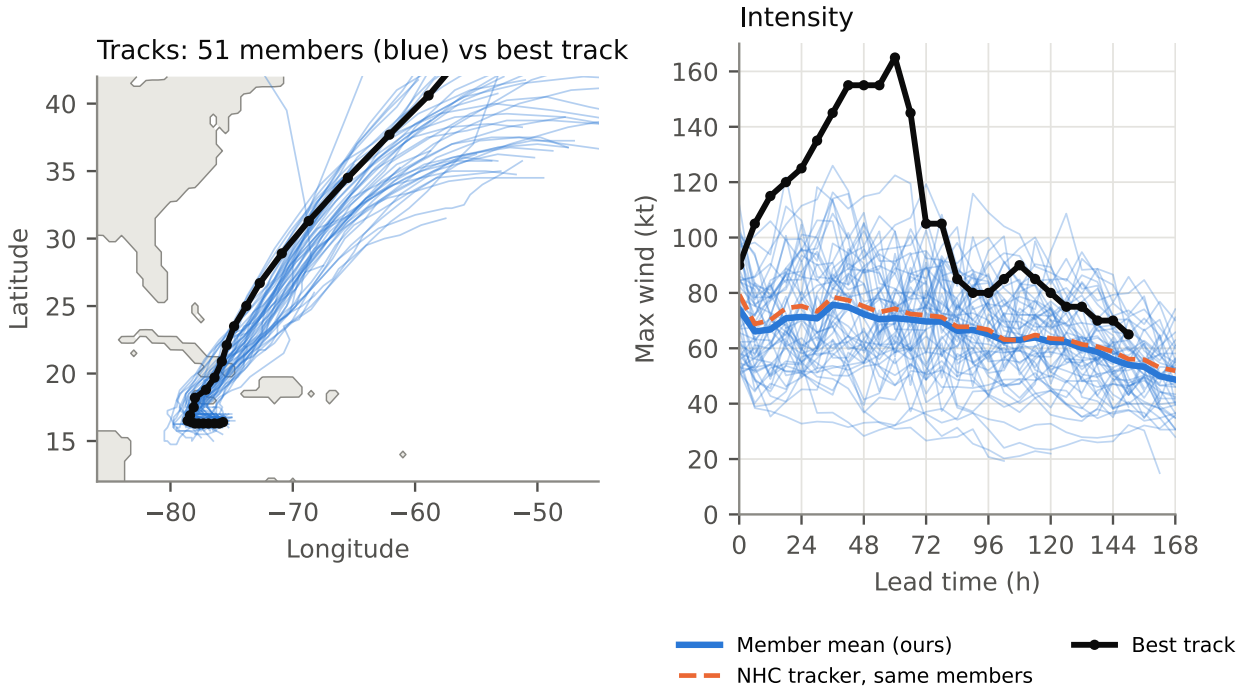


Figure 5: IFS-ENS run of 26 October 2025 for Hurricane Melissa: our member tracks (left) and member intensities (right) against the best track. At 0.25° the ensemble cannot represent the storm's inner core; the ensemble-mean intensity stays near 70–75 kt while Melissa reached 165 kt.

were run on identical cases with leave-one-season-out cross-validation: correcting the ensemble's own mean intensity from gridded features alone (as AIFS-TC did for AIFS), the consensus correction retrained on the same sample, and the consensus correction with the gridded features added. Gridded features alone are not enough: they remove much of the ensemble's bias but remain 2–3 kt behind HCCA. Added to the consensus correction, the IFS-ENS features reduce the error by about 1.0 kt over 2024–2026 and by about 1.5 kt over 2025–26, both significant, and by about 2 kt on rapid intensification in 2025–26. The gain is larger than in earlier versions because the consensus no longer sees any ECMWF aids, so our own IFS-ENS extraction is now its only source of ECMWF information. The GEFS features add nothing (−0.02 kt over 2020–2026, with a confidence interval of ±0.2 kt), even after the GEFS archive was extended back to October 2020 to give five seasons of training data instead of two. The likely reason is that NHC's tracker output for the GEFS members is already part of the consensus inputs.

6 Calibrated intensity ensembles

6.1 Construction

The raw member intensities are weak and under-dispersive: in 2025–26 their 10–90% ranges contain only about 20–40% of outcomes and their spread is a quarter to a third of their actual error. Correcting each member individually with a learned model turned out to collapse the ensemble, because the model learns to send every member to nearly the same corrected value. The final construction instead separates *where* the ensemble is centred, *how wide* it is, and *how its members differ*, and takes each from the source that does it best.

Shape from the physical ensemble. For member i at lead τ we form its departure from the ensemble mean, re-anchored so that every member starts at zero at t_0, and smooth it in time with a Gaussian kernel of 6 h:

$$d_i(\tau) = [v_i(\tau) - \overline{v}(\tau)] - [v_i(0) - \overline{v}(0)], \quad \tilde{d}_i = \mathrm{smooth}(d_i), \quad z_i(\tau) = \tilde{d}_i(\tau) / \ \mathrm{sd}(\tau),$$

where $\mathrm{sd}(\tau)$ is the (lead-smoothed) standard deviation of the smoothed departures within that forecast, so that z_i carries only the ensemble's scenario structure. The 6-h kernel was chosen because it makes the members' step-to-step behaviour match the best track's (section 6.4) while costing nothing in skill; wider kernels made the members unrealistically smooth.

Size from a spread model. A separate LightGBM model predicts, for each forecast and lead time, the expected absolute error $\hat{\sigma}$ of the corrected consensus. Its inputs are the consensus-correction features (the disagreement between the early aids, the spread of NHC's GEFS member tracks, the storm's intensity, recent trend and position, the lead time) together with the width of the consensus correction's own quantile range and its predicted 24-hour intensity change. It is trained on the consensus correction's out-of-fold errors from 2018–2024, so it learns from eight seasons of realistic forecast errors rather than from the two seasons of gridded ensembles. On 2025–26 its prediction correlates with the actual error at 0.4 (0.33 within a fixed lead time), and it ranks forecasts well (Table 3).

Cases binned by predicted error	Cases	Predicted error (kt)	Actual error (kt)
Calmest fifth	1650	5.0	5.3
Second	1649	7.9	9.0
Middle	1649	11.0	13.6
Fourth	1649	14.6	15.6
Most volatile fifth	1650	21.0	19.0

Table 3: 2025–26 forecasts binned by the spread model's predicted error, against the corrected consensus's actual mean absolute error.

Centre from a consensus. The members are placed around a consensus mean $\hat{v}_c$:

$$\mathrm{member}_i(\tau) = \hat{v}_c(\tau) + s \cdot k(\tau) \cdot \hat{\sigma}(\tau) \cdot z_i(\tau),$$

where $\hat{\sigma}$ is smoothed along lead time and $k(\tau)$, a slowly varying factor of about 1.2–1.6 that converts a mean absolute error into a member spread, is fitted by minimising the continuous ranked probability score (CRPS) on out-of-fold seasons.

For the GEFS product, **GECO**, the centre is NHC's corrected consensus HCCA exactly as issued, and $s = 1$ (the CRPS-fitted spread). HCCA is in NHC's public real-time files at every cycle (0.3% of 2025–26 forecasts lacked it and fall back to our own consensus), and it draws on the ECMWF guidance that NHC has internally but our public-data consensus cannot use. Centring on HCCA rather than on our consensus correction improved the CRPS by −0.34 kt [−0.61, −0.08] over nine seasons and by −0.71 kt [−1.62, 0.1] over 2025–26, mostly in 2026. HCCA is used as issued; smoothing it along lead time, as our own consensus is, cost 0.13 kt. Trimming the spread to 80% of the fitted value cost about 0.1 kt and lowered the coverage, so GECO keeps the full spread. The fitted factor is $k = 1.2$ at all leads.

The IFS-ENS product, **IECO**, is built in exactly the same way from our 51 IFS-ENS members (00 UTC run, used at 12 UTC), with $k = 1.5 - 0.3\tau/100$ fitted on 2024, the only out-of-fold IFS-ENS season. An earlier version centred IECO on the consensus correction with IFS-ENS gridded features (section 5); on the same 2025–26 cases it was 0.5 kt worse than the HCCA-centred version and no better than dressed HCCA (`docs/ensemble_ifs_hcca.md`). On identical 12 UTC cases IECO is 0.1 kt behind GECO (significant), probably because its run is 12 h old against 6 h for NHC's GEFS tracks, and pooling the two ensembles around HCCA gained nothing.

Each step fixed a concrete problem seen along the way. Scaling raw member departures independently at every lead time produced traces that zig-zagged by tens of knots; anchoring and smoothing removed that. A spread that depended only on lead time was too wide for calm storms and too narrow for volatile ones; the spread model fixed that. The ECMWF ensemble's raw spread roughly doubled between 2024 and 2025, probably because of a model upgrade, which made any scaling learned on earlier seasons far too wide; because the spread now comes from the consensus error model and the raw ensemble only contributes its shape, such

changes no longer affect the spread. An attempt to make the spread asymmetric (wider above the centre than below in rapid-intensification situations) made the ensembles worse and is not used.

6.2 Nine seasons of GECO

Because GECO needs only NHC's track files, it can be evaluated on every season since 2018 (336 storms, 27623 cases, all four cycles, leads 12–120 h). For 2018–2024 every fitted ingredient is out of sample for the season being scored: the spread model and $k(\tau)$ are fitted without that season. The fair yardstick for a probabilistic forecast is a probabilistic version of the deterministic consensus: "HCCA dressed" turns HCCA into a normal distribution whose width at each lead is HCCA's root-mean-square error in the other seasons. (Scoring HCCA as a single value would flatter any ensemble, whose CRPS is always below its own mean's absolute error.) GECO beats dressed HCCA in all nine seasons, each significantly, by −0.36 kt [−0.41, −0.29] overall (Table 4). The gain is steady across the change from the 20-member GEFS to the 30-member GEFSv12 in September 2020 and the switch from HWRF to HAFS in 2023, and the refitted $k(\tau)$ stayed at 1.2–1.3. GECO's ensemble mean is HCCA itself, so the gain comes entirely from the GEFS members' scenario shapes and the storm-specific spread. Our own consensus as the centre, with the same members and spread, is level with dressed HCCA over 2018–2024 but falls behind it in 2026.

Season	Storms	Raw GEFS	Consensus centre	GECO	HCCA dressed	OFCL dressed	GECO − HCCA dressed	GECO coverage
2018	41	15.64	7.73	7.60	7.99	8.10	−0.39 [-0.54, -0.20]	73%
2019	37	15.67	7.56	7.38	7.81	8.08	−0.43 [-0.59, -0.28]	73%
2020	52	12.20	6.93	6.73	7.16	7.22	−0.43 [-0.60, -0.25]	74%
2021	40	9.89	7.08	6.60	6.97	7.11	−0.37 [-0.54, -0.23]	74%
2022	33	9.85	6.60	6.44	6.73	7.02	−0.28 [-0.50, -0.06]	84%
2023	39	10.87	6.72	6.58	6.85	6.98	−0.30 [-0.49, -0.11]	86%
2024	32	14.28	7.89	7.44	7.75	7.94	−0.31 [-0.49, -0.14]	73%
2025	30	15.57	7.64	7.77	8.08	7.59	−0.32 [-0.53, -0.13]	74%
2026	32	13.94	8.19	6.90	7.23	7.74	−0.33 [-0.45, -0.21]	77%
2018–24	274	12.69	7.21	6.97	7.33	7.49	−0.36 [-0.43, -0.29]	77%
2025–26	62	14.61	7.96	7.26	7.58	7.68	−0.32 [-0.43, -0.20]	76%
all	336	13.10	7.37	7.04	7.39	7.53	−0.36 [-0.41, -0.29]	76%

Table 4: Fair CRPS (kt) of NHC's GEFS member tracks by season, identical cases. Raw GEFS: members offset to the observed intensity at t_0 (as NHC's interpolation does). Consensus centre: the same construction around our own consensus correction. HCCA / OFCL dressed: each as a normal distribution with its out-of-season error by lead. The difference column has 95% storm-bootstrap intervals; coverage is the share of outcomes inside GECO's 10–90% range (ideal 80%).

6.3 Probabilistic skill compared with FNV3

Ensemble	CRPS (kt)	Bias of mean (kt)	10–90% coverage	Spread / error
GEFS: raw members	22.6	−25.5	22%	0.2
GEFS: GECO	7.6	−1.2	75%	0.87
GEFS cycles: FNV3 members	8.4	−6.0	62%	0.69
IFS-ENS: raw members	18.2	−22.5	41%	0.4
IFS-ENS: IECO	7.7	−1.1	80%	0.95
IFS-ENS cycles: FNV3 members	8.3	−5.9	62%	0.69

Table 5: 2025–26, leads 12–120 h, identical cases within each block (GECO: 61 storms, all cycles; IECO: 56 storms, 12 UTC cycles). FNV3 members are taken from the newest run at least 6 h old, the same timing rule as ours. Spread / error is the mean member spread divided by the root-mean-square error of the ensemble mean (1 for a well-dispersed ensemble).

Over 2025–26 GECO beats FNV3′s members: its CRPS is lower by −0.78 kt [−1.6, −0.01], a significant difference, by −0.41 kt [−2.23, 1.02] in 2025 and −1.02 kt [−1.82, −0.14] in 2026, with a mean bias of about −1 kt against FNV3′s −6 kt (Table 5). It is better than or level with FNV3 at every lead to 120 h, most clearly in the first two days, and is also ahead of FNV3′s late run, which arrives too late for the forecast cycle (−0.56 kt, not significant).

IECO is also ahead of FNV3′s members on the 12 UTC cycles: −0.65 kt [−1.53, 0.2], in 2025 (−0.36 kt [−2.08, 1.16]) and in 2026 (−0.83 kt [−1.71, 0.06]), though not significantly. The raw ensembles are more than twice as bad. FNV3′s members are sharp but over-confident in both seasons: their spread is about 60–70% of their error and their mean is 5–6 kt too weak. The spread of both GECO and IECO is close to their error (Figure 6); in both, the spread widens in situations the spread model recognises as volatile.

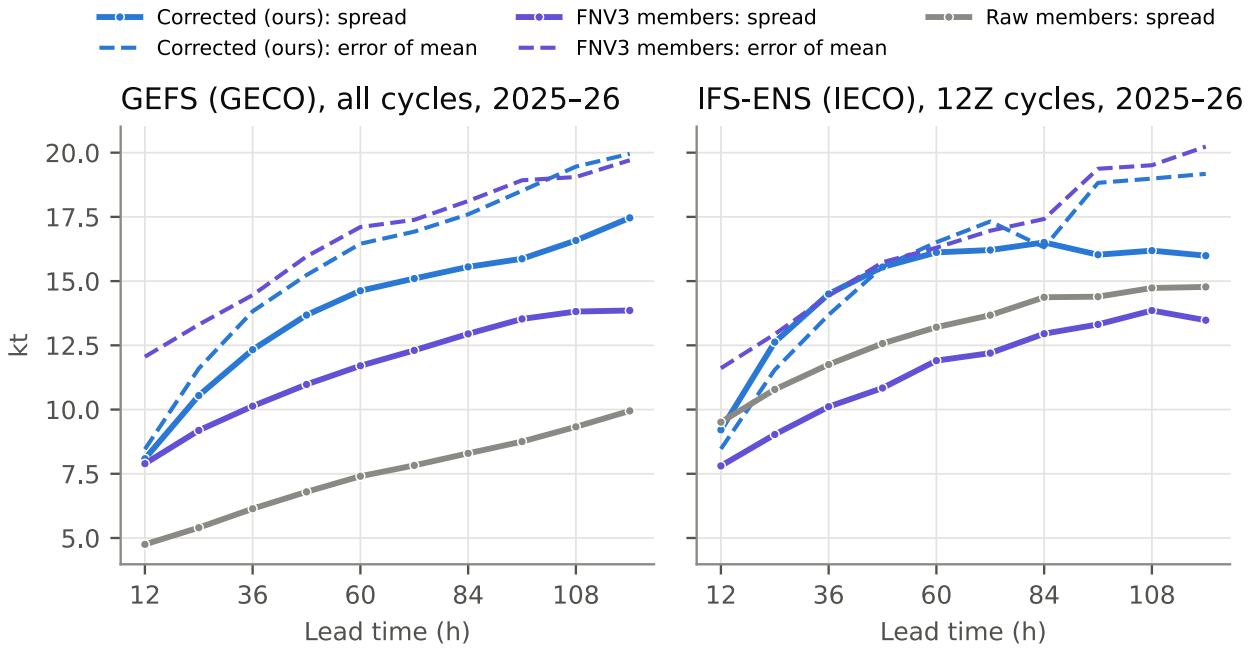


Figure 6: Mean member spread (solid) and the error of the ensemble mean (dashed) by lead time, 2025–26. A well-calibrated ensemble has the two curves close together. The raw ensembles' errors (25–40 kt) are off the scale and only their spread is shown.

6.4 Probability of rapid intensification

Counting the share of members that intensify by 30 kt in 24 hours is a poor way to estimate the probability of rapid intensification from these ensembles, because it depends on exactly how the members are smoothed and scaled. Instead a separate LightGBM classifier, trained on 2018–2024 with the same early-cycle features plus the corrected consensus's own predicted 24-hour change, predicts the probability that the best track will rise by at least 30 kt in the 24 hours ending at each lead time; an isotonic recalibration fitted on out-of-fold predictions keeps its probabilities honest. On 2025–26 it is 19% better than climatology, a little behind FNV3′s members (23%), and ranks cases as well (Table 6). Both systems under-forecast: 2025–26 was rich in rapid intensification (9% of cases against 6% in 2018–2024), and at most probability levels the observed frequency is higher than forecast (Table 7). For the standard "next 24 hours" product the model's Brier score is 0.068 against 0.093 for climatology. Work on the RI probability is paused, since it is a side product rather than a correction of the physical ensembles.

Probability of rapid intensification	Brier score	Skill vs climatology	AUC
Climatology (2018–24 rate)	0.084	0%	–
RI model (calibrated)	0.068	19%	0.87
FNV3 members (share rising ≥ 30 kt / 24 h)	0.065	23%	0.86

Table 6: Probability of a ≥ 30 kt rise in the 24 h ending at $t_0 + \tau$, τ = 24–120 h, 2025–26 (5342 cases, 490 with rapid intensification). Skill is the reduction in Brier score relative to climatology; AUC measures ranking only.

Forecast probability	Cases	Mean forecast	Observed frequency
[0.0, 0.05)	4039	0.018	0.026
[0.05, 0.1)	324	0.064	0.139
[0.1, 0.2)	511	0.15	0.249
[0.2, 0.3)	171	0.239	0.444
[0.3, 0.5)	160	0.377	0.45
[0.5, 1.0)	137	0.59	0.489

Table 7: Reliability of the RI model's probabilities on 2025–26: forecasts in each probability band against the observed frequency.

6.5 Physical consistency

Traces	Zig-zag rate	Mean |12-h change| (kt)	Share of 12-h changes > 30 kt
Best track	0.034	8.6	1.9%
FNV3 members	0.070	9.9	4.2%
GEFS raw members	0.014	5.1	0.3%
IFS-ENS raw members	0.032	6.5	1.0%
GEFS first corrected version	0.088	11.3	5.5%
IFS-ENS first corrected version	0.103	12.5	8.3%
GECO (GEFS)	0.025	9.0	2.1%
IECO (IFS-ENS)	0.023	8.9	2.5%

Table 8: Step-to-step behaviour of intensity traces, 2025–26. A zig-zag is two consecutive 12-h changes of opposite sign, each at least 10 kt.

Statistics alone do not show whether individual members look like plausible storms, so we also measured how traces evolve from one time step to the next (Table 8). The final corrected members change a little less from step to step than the best track does, reverse direction less often, and rarely change by more than 30 kt in 12 hours. The first corrected version, by contrast, zig-zagged two to three times as often as real storms. FNV3's members have a 12-hour zig-zag rate similar to the best track's, but at their native 6-hour spacing they reverse direction about 21 times as often as the best track (0.103 against 0.005), which is the short-term jitter visible in their traces.

6.6 Case studies

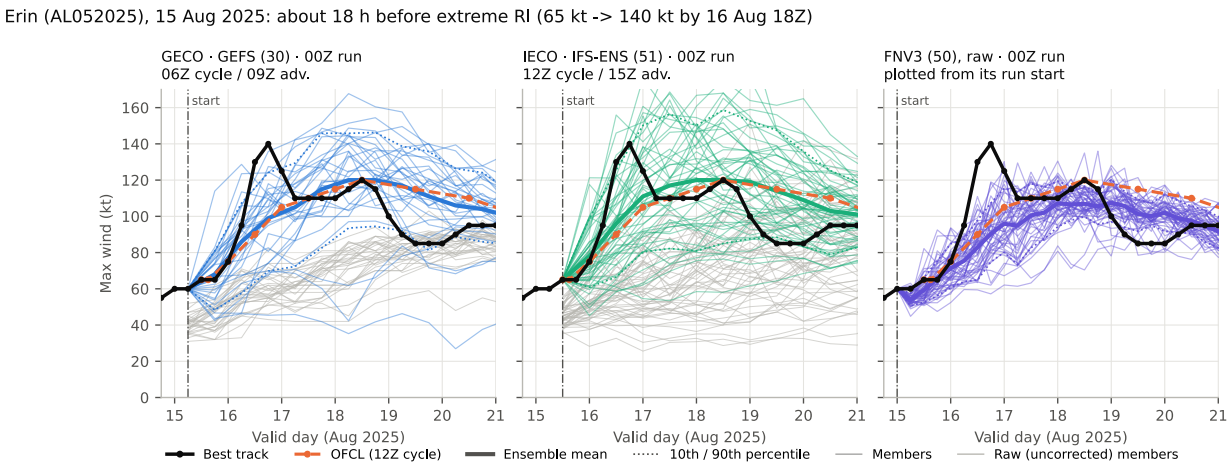


Figure 7: Hurricane Erin, 15 August 2025, about 18 h before its extreme rapid intensification (65 kt to 140 kt by 16 August 18Z): GECO (NHC's 30 GEFS member tracks) for the 06Z cycle (09Z advisory, from the 00Z run), IECO (our 51 IFS-ENS members) for the 12Z cycle (15Z advisory, from the 00Z run), FNV3's 00Z members, the best track and the official forecast of the 12Z cycle. In grey, the raw members, starting at the models' own intensity.

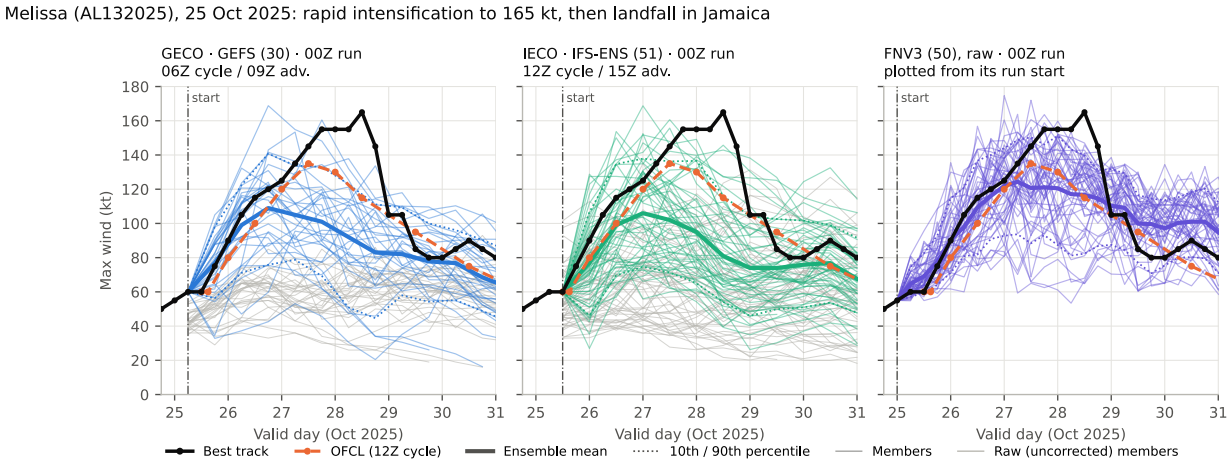


Figure 8: Hurricane Melissa, 25 October 2025, at the start of its rapid intensification to 165 kt.

The case studies show what the numbers mean in practice. For Erin on 15 August (Figure 7) the spread model recognises a volatile situation: both corrected ensembles fan out widely in the first day, and their upper members reach the 140-kt peak, whereas FNV3′s members stay mostly below about 125 kt and the official forecast peaks near 120 kt. For Melissa (Figure 8) the corrected ensembles again rise quickly from the observed intensity and fan out; their means peak well below the 165-kt best track, as the consensus does, but the upper members reach 140–160 kt, so the extreme outcome lies within the ensemble. FNV3′s mean comes closer to the peak and its upper members also reach about 160 kt. Eight further cases are in Appendix C, including a steady storm where the spread stays narrow, two forecasts made days before a rapid intensification and an East Pacific storm that decays over cooler water. A larger set of 28 cases, with several start times for Erin, Gabrielle, Humberto, Melissa and Kiko, is collected in a separate file (`docs/case_plots.pdf`). In these figures each system starts at a different time, following the early-cycle rule. A second file (`docs/case_plots_12z.pdf`) shows the same cases with all three for the same 12Z cycle (GECO from the 06Z GEFS run, as it would run in real time, IECO from the 00Z IFS-ENS run and FNV3 from its 06Z run) for a like-for-like visual comparison.

6.7 Remaining limitations

The ensembles are symmetric around a consensus that, like every guidance system, under-predicts rapid intensification, so in such situations the truth still falls above the 90th percentile more often than it should, and some lower members describe weakening that did not happen. Making the spread asymmetric did not help, because which borderline situations will intensify rapidly cannot be identified well enough in advance. Both products inherit HCCA's errors, which they cannot improve on in the mean. The RI probability is reliable in ranking but too low in RI-rich seasons. Three follow-up attempts to fix the RI shortfall, documented in `docs/ri_improvements.md` and `docs/ensemble_ritail.md`, had limited success. Adding SHIPS's environmental predictors to the RI model did not help. Adapting its base rate to the RI frequency seen earlier in the same season did not help either, because a season's early part says little about the rest. Stretching the ensemble's upper half in proportion to the RI probability helped only marginally. Feeding NHC's own operational RI guidance (SHIPS-RII, DTOPS and related probabilities, already in the SHIPS files) into the RI model gave a small but significant improvement. Because the RI probability is a side product rather than a correction of the physical ensembles, further work on it is paused.

7 Conclusions and next steps

For physical-model guidance, a learned early-cycle correction built only from NHC's public real-time files reaches the operational frontier over 2018–2025 and improves on it during rapid intensification. The 2026 season, the first run exactly as in real time, shows its limit: without the ECMWF guidance that NHC uses inter-

nally it falls about 2 kt behind HCCA. The gridded IFS-ENS fields are the real-time route to that information, and they now reduce the consensus error by 1–1.5 kt; the GEFS fields add nothing, even with five seasons of training data. The calibrated ensemble combines the accuracy of the corrected consensus, a storm-specific estimate of its uncertainty and the scenario structure of the physical ensembles. GECO, NHC's GEFS member tracks reshaped around HCCA, beats a probabilistic HCCA in each of nine seasons and FNV3′s members over 2025–26, at every cycle and from public real-time files only. IECO, the same construction for the IFS-ENS, is a little behind GECO but also ahead of FNV3′s members. Neither uses gridded data in its centre or spread: further gains from the model fields would probably need a convolutional network on the full gridded fields, as in AIFS-TC.

A multi-model ensemble that pools the scenario shapes of NHC's GEFS and ECMWF member tracks around the same corrected centre was slightly better than either model alone in 2025 (CRPS 7.98 against 7.99–8.06 kt, not significant) and missed the upper tail less often in rapid-intensification runs. Doing the same with our own GEFS and IFS-ENS members at a common start time (GEFS used 12 h old) gave no measurable gain over 2025–26 (−0.03 kt): once both ensembles share the corrected centre and the predicted spread, their scenario shapes are close to interchangeable.

The aim of the project is the best possible correction of the physical ensembles, not the best forecast overall, so FNV3 remains a benchmark and is not used as an input. The goal is a real-time system; using an older IFS-ENS run in the early window is acceptable. The GEFS archive has been extended back to October 2020 (492 runs), which settled that GEFS fields add nothing, so the GEFSv12 reforecasts are no longer a priority. The next steps, in order, are:

- Extend IECO to the 00 UTC cycle from the 12 UTC IFS-ENS run.
- A convolutional network on multi-level gridded fields, combined with the consensus correction, when compute allows.
- Re-examine the 2026 shortfall once NHC finalises the 2026 best tracks, and look at which storms and lead times drive it.
- A real-time prototype triggered by NHC's public guidance files, after checking when each data source becomes available relative to the early-cycle deadline.

Appendix A · Verification framework

All verification follows NHC's conventions. Truth is the final best track (b-deck); a case counts only if the system is a tropical or subtropical cyclone at both the initial and the valid time; and every comparison is homogeneous, i.e. all systems are scored on exactly the same cases. Deterministic forecasts are scored by the mean absolute error and bias of maximum wind; ensembles by the fair continuous ranked probability score (CRPS), a quantile score at the 10th, 50th and 90th percentiles, the coverage of the 10–90% range and the ratio of spread to the error of the mean. Confidence intervals resample whole storms (block bootstrap), as in the AIFS-TC paper, because cases from the same storm are strongly correlated. Rapid intensification is an observed increase of at least 30 kt over the 24 hours ending at the valid time.

As a check, the harness recomputes NHC's published 2024 Atlantic verification. The official-forecast table is reproduced with identical case counts and errors within 0.1 kt, and the homogeneous early-model table with identical case counts and 94 of 96 errors within 0.2 kt (Figure 9). The AIFS-TC paper's 2025 values for OFCL and FNV3 are reproduced within their confidence intervals.

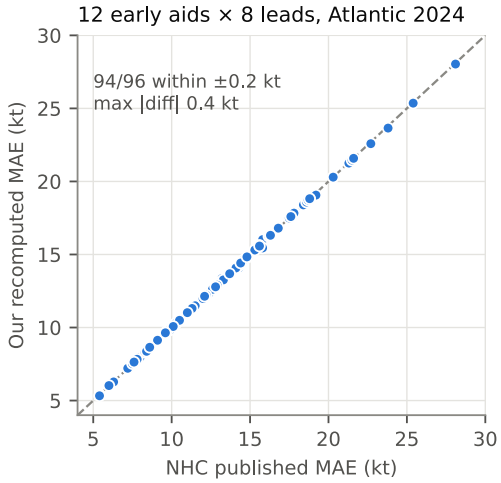


Figure 9: Recomputed versus published errors for NHC's 2024 homogeneous early-aid verification; each point is one aid at one lead time.

Appendix B · Detailed results

All values in kt; identical cases across the systems in each table; 95% storm-bootstrap intervals.

Consensus correction, cross-validation 2018–24

System	MAE	95% CI	Bias
Ours, no SHIPS (TCCA)	10.2	9.5–10.9	0.5
Ours, + SHIPS (TCCB)	10.2	9.5–10.9	0.5
HCCA	9.9	9.3–10.5	−0.8
IVCN	10.0	9.2–10.7	−2.5
OFCL	10.0	9.3–10.7	0.1
Uncorrected aid mean	11.1	10.2–11.9	−2.6

Consensus correction, held-out 2025–26

System	MAE	95% CI	Bias
Ours, no SHIPS (TCCA)	11.4	9.8–13.0	−2.2
Ours, + SHIPS (TCCB)	11.3	9.7–12.9	−2.2
HCCA	10.3	9.1–11.5	−0.9
IVCN	11.7	9.9–13.3	−6.0
OFCL	10.3	9.0–11.7	−0.1
Uncorrected aid mean	12.7	10.7–14.6	−6.0

Rapid intensification, 2018–24 (86 storms)

System	MAE	95% CI	Bias
Ours, no SHIPS (TCCA)	19.9	17.6–22.0	−17.5
Ours, + SHIPS (TCCB)	20.1	17.8–22.4	−18.0
HCCA	21.6	19.5–23.6	−20.2
IVCN	25.7	23.6–27.9	−25.2
OFCL	21.7	19.5–23.9	−20.6

Rapid intensification, 2025–26 (22 storms)

System	MAE	95% CI	Bias
Ours, no SHIPS (TCCA)	19.2	15.1–23.5	−16.6
Ours, + SHIPS (TCCB)	19.6	15.6–23.5	−17.0
HCCA	19.3	14.9–23.9	−17.7
IVCN	27.9	24.5–31.4	−27.8
OFCL	19.3	15.7–23.6	−18.4

System	MAE	95% CI	Bias
Uncorrected aid mean	30.4	28.4–32.3	−30.2

System	MAE	95% CI	Bias
Uncorrected aid mean	32.8	29.2–36.1	−32.7

Error by lead, held-out 2025–26

System	12	24	36	48	60	72	96	120
Ours, no SHIPS (TCCA)	6.8	9.3	11.2	12.8	14.0	14.1	14.7	15.8
Ours, + SHIPS (TCCB)	6.9	9.2	11.1	12.7	13.8	14.0	14.5	15.4
HCCA	5.8	8.3	10.3	11.3	12.3	12.6	13.9	15.7
IVCN	6.0	8.9	11.3	12.9	14.4	15.0	16.7	18.1
OFCL	5.2	8.1	10.2	11.7	12.6	13.0	14.4	16.2
Uncorrected aid mean	6.3	9.7	12.2	13.9	15.3	16.4	19.0	20.8

Error by lead against FNV3 early, 2025–26, 12–168 h (56 storms)

System	12	24	36	48	60	72	84	96	108	120	132	144	156	168
Ours, no SHIPS (TCCA)	7.0	9.5	11.3	13.1	14.1	14.8	15.3	15.5	15.7	15.9	14.9	17.3	16.3	14.8
Ours, + SHIPS (TCCB)	7.0	9.4	11.3	13.0	13.9	14.6	15.2	15.2	15.2	15.4	15.0	17.6	15.9	14.2
FNV3 early (GDMI)	6.0	8.4	10.1	11.6	12.7	13.7	14.3	15.2	15.9	16.8	16.4	15.4	14.2	13.8
HCCA	5.9	8.5	10.4	11.6	12.4	12.6	13.0	13.8	14.9	15.7	17.1	17.6	17.8	17.6

Gridded features (leave-one-season-out, leads 12–120 h; from `docs/m2_results.md`). On IFS-ENS cycles (2024–2026): consensus correction retrained on the same rows 12.0 kt, with IFS-ENS gridded features 10.9 kt (difference −1.04 kt [−1.72, −0.40]), gridded features alone 13.4 kt, HCCA 10.4 kt, OFCL 10.4 kt. On GEFS cycles (2020–2026): 10.41 kt without and 10.39 kt with GEFS gridded features (−0.02 kt [−0.22, +0.18]), HCCA 9.6 kt. **Pressure (2025–26, 12–120 h):** corrected 8.8 mb against 12.3 mb for the reference and 17.1 mb for persistence.

Appendix C · Additional case studies

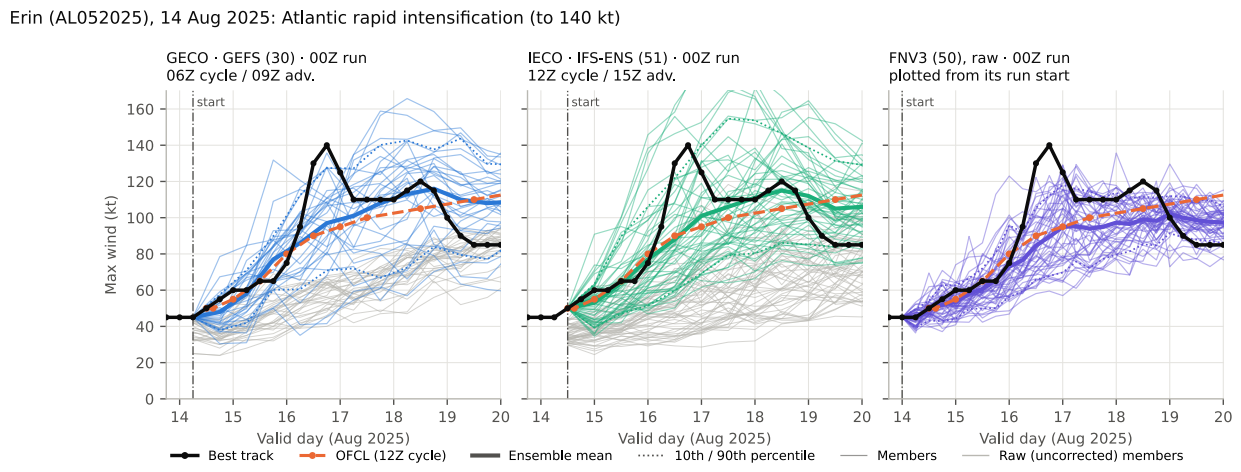


Figure 10: Hurricane Erin, 14 August 2025, two days before its rapid intensification.

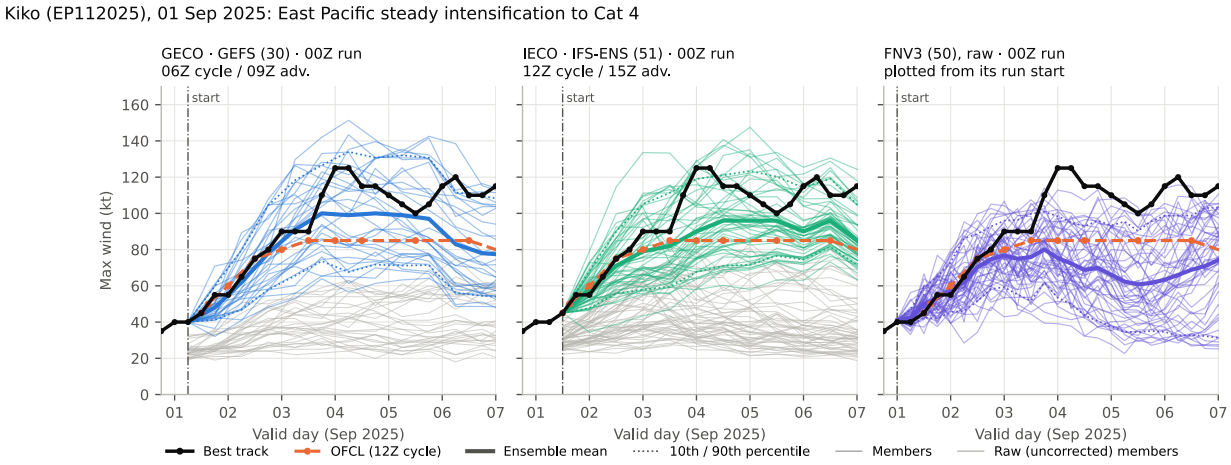


Figure 11: Hurricane Kiko, 1 September 2025, East Pacific: our ensembles bracket the best track while FNV3's remain far too weak after the second day.

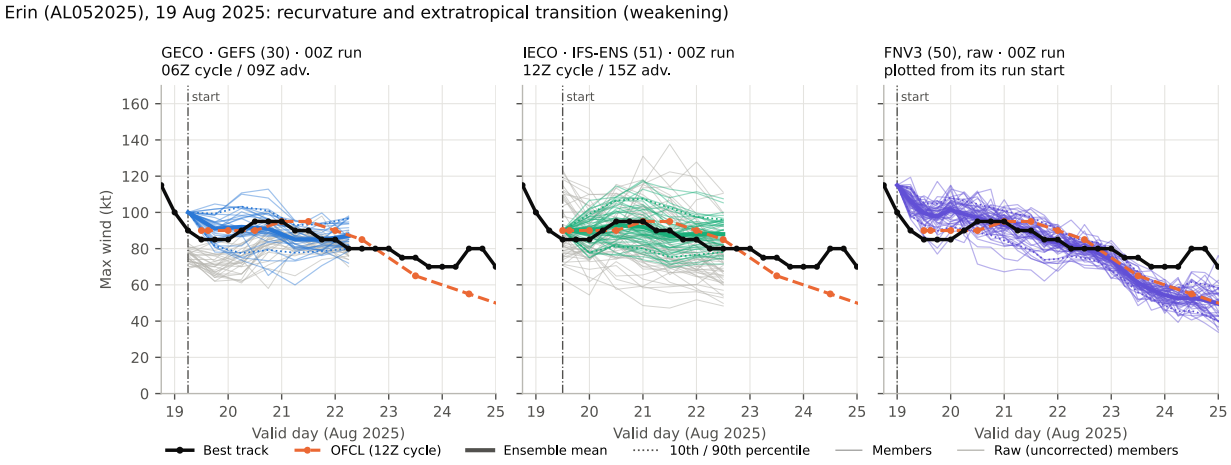


Figure 12: Hurricane Erin, 19 August 2025: recurvature and extratropical transition.

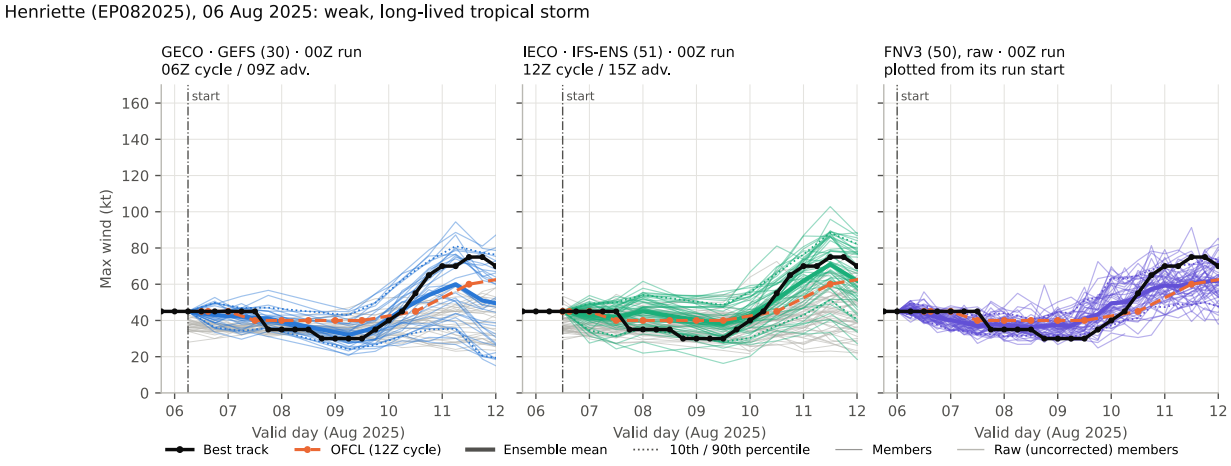


Figure 13: Tropical Storm Henriette, 6 August 2025: a weak, long-lived storm.

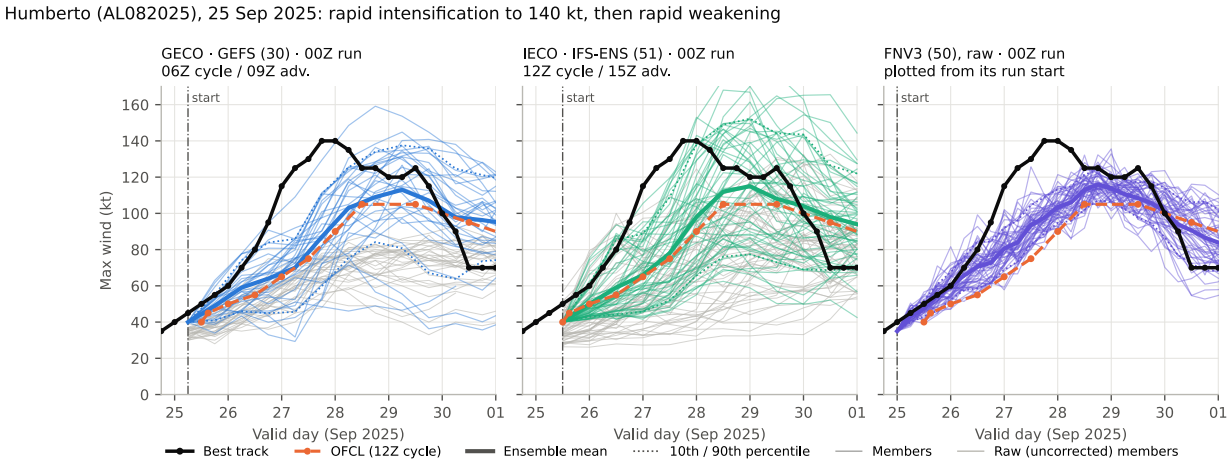


Figure 14: Hurricane Humberto, 25 September 2025: rapid intensification to 140 kt within three days. All three ensembles are too slow; only our upper members reach the peak, a day late.

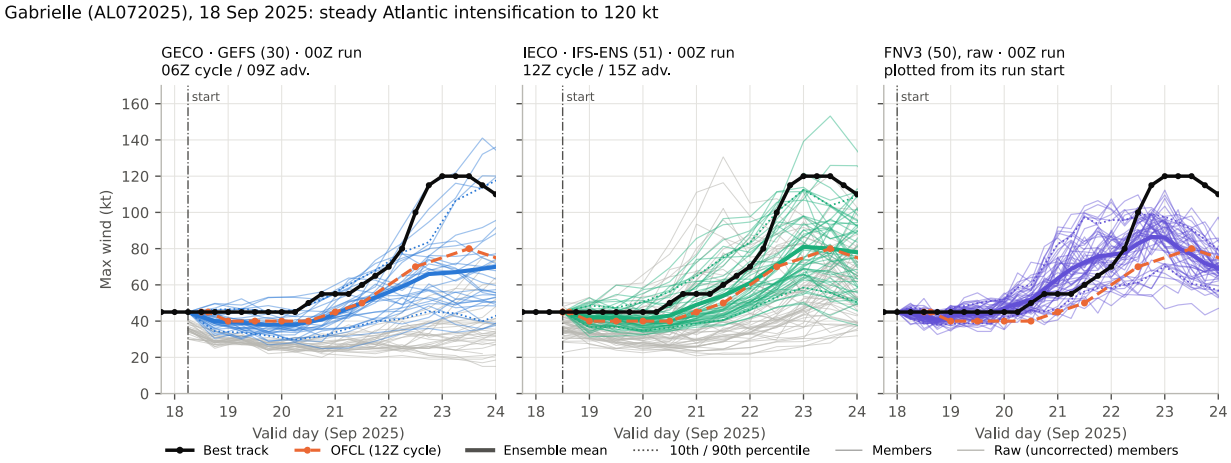


Figure 15: Hurricane Gabrielle, 18 September 2025: weak for four days, then rapid intensification to 120 kt. Every system misses the timing and size of the burst; FNV3's mean comes closest at the peak.

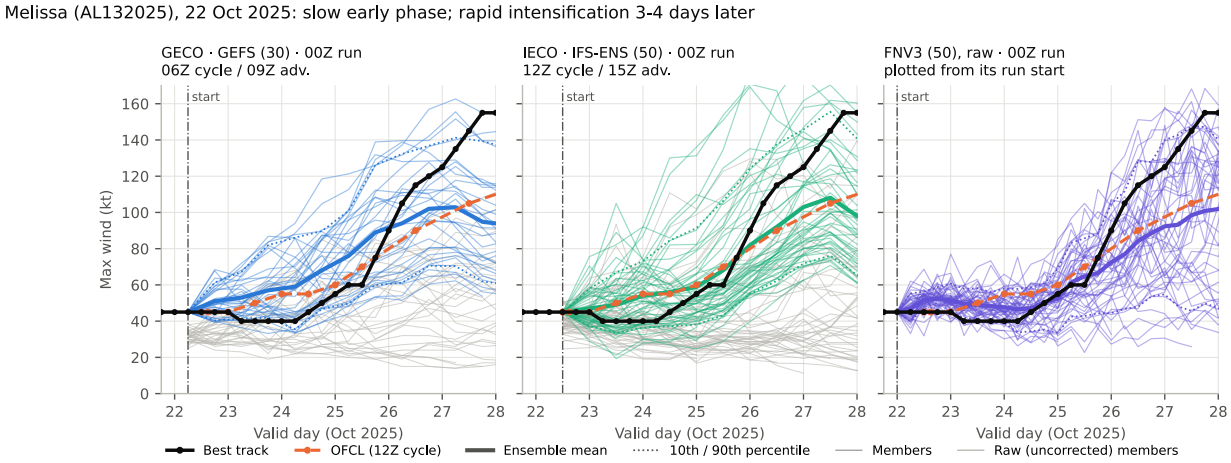


Figure 16: Hurricane Melissa, 22 October 2025, three to four days before its rapid intensification. Our ensembles already lean toward a strong hurricane and keep the best track inside their range; FNV3 stays weak until day four.

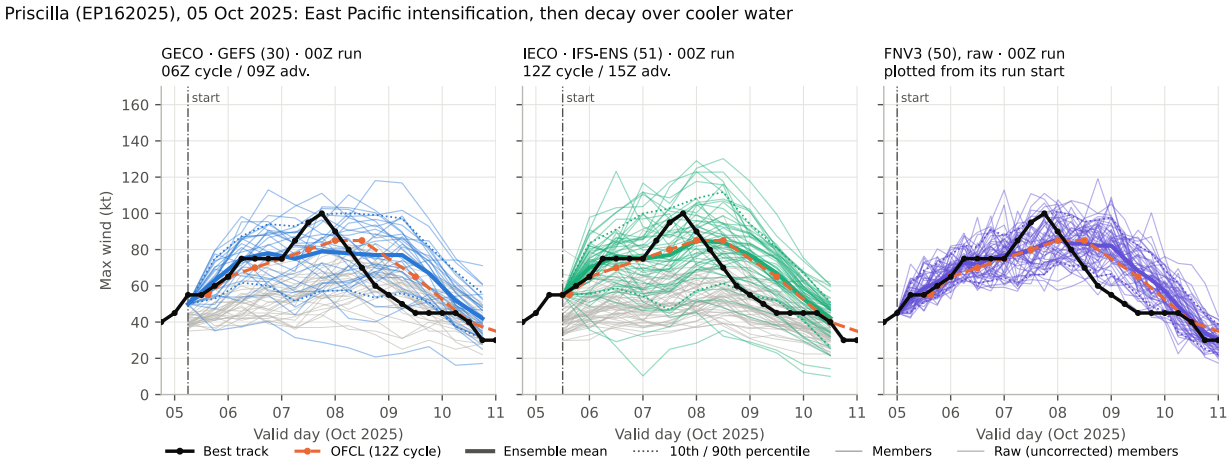


Figure 17: Hurricane Priscilla, 5 October 2025, East Pacific: rise to 100 kt and decay over cooler water. All systems capture the shape; ours peak lower and decay later than the best track.

Appendix D · Data coverage

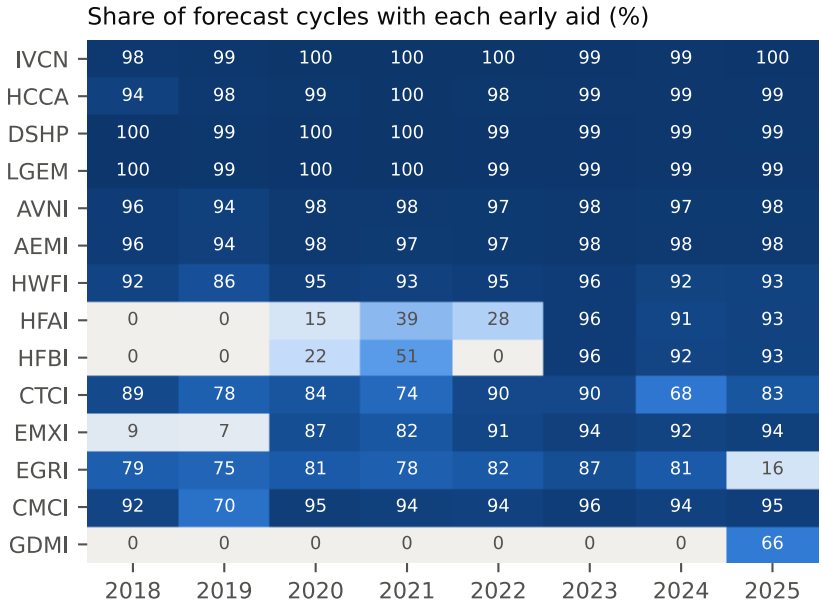


Figure 18: Share of NHC forecast cycles with each early aid, after merging the 6-h and 12-h interpolated versions. HAFS (HFAI/HFBI) became operational in 2023; ECMWF aids appear in the public archive from 2020 but never in the real-time files; FNV3 (GDMI) only from 2025.

The operational SHIPS archive covers 70–96% of cases per season (lowest in 2020–21) and reaches 168 h only from 2020. GEFS members are available for nearly all cases; ECMWF members for about three quarters of archive cases from 2020 and very few before, and none in real time. Our own extraction covers every 00 UTC run with an active storm: IFS-ENS from June 2024 and GEFS from October 2020, both to October 2026 (1,186 runs). FNV3 members for 2025–26 come from Google DeepMind's Weather Lab archive (50 members, CC BY 4.0 for historical data); their ensemble mean matches the GDMN aid in NHC's files.

Appendix E · Technical notes and limitations

Data access. Subsetting a cloud dataset does not always reduce what is downloaded: on dynamical.org's "virtual" stores each stored chunk is a whole global field, so a storm-centred patch costs as much as the globe, whereas the "native" stores are tiled and make patches cheap but hold only surface variables. The gridded extraction therefore reads per-member surface fields from the native stores and pressure-level environment fields from one member only (the GEFS virtual store and ECMWF open data). Reading the zarr arrays directly, rather than through xarray's lazy indexing, cut memory use by a factor of about four.

Running on a shared machine. The extraction ran on a shared DGX Spark, capped at four cores and 16 GB, at the lowest CPU and I/O priority, with downloads limited to 85% of the network link minus other users' traffic and automatic pauses when the machine was busy. Work was split into one unit per model run, written atomically, with a lock-file heartbeat so that a crash or reboot loses at most one unit and the next start resumes; failed units are retried. The original pilot (about 0.75 TB downloaded, 6 GB stored) took about 20 hours of network-bound work; the GEFS extension back to October 2020 (492 runs, about 210 GB) took 7.5 hours overnight. Our tracker's agreement with NHC's tracker over all units is shown in Figure 19.

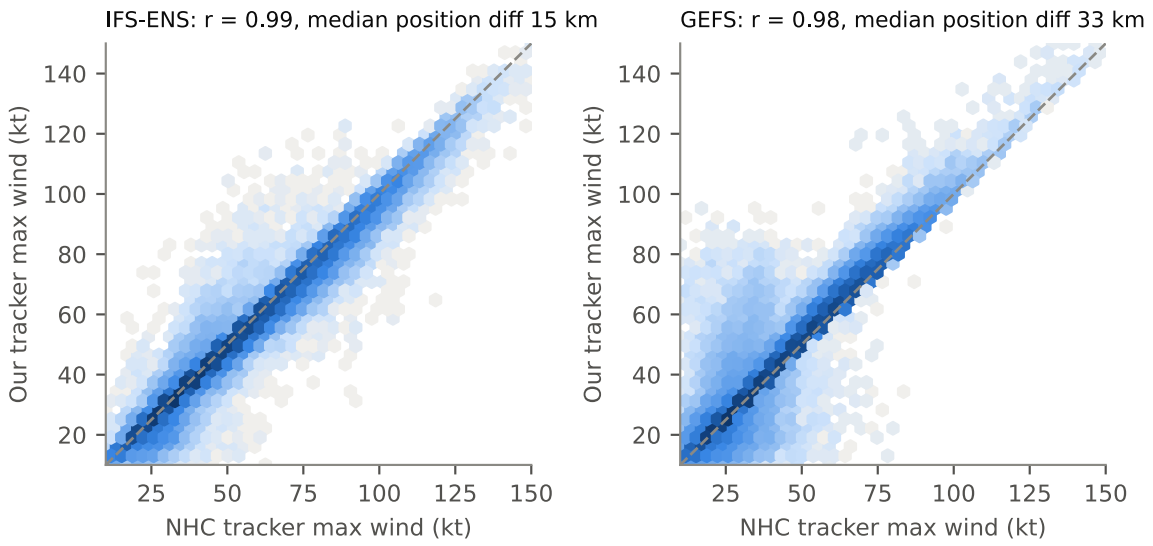


Figure 19: Our tracker's maximum wind against NHC's tracker on the same members and runs.

Real-time availability. ECMWF's track aids (EMX, EEMN and the members) are absent from NHC's public real-time files during the season and appear in the archive only afterwards, so the consensus correction is trained without them; our own extraction from ECMWF open data is the legitimate real-time route. From mid-May 2026 ECMWF's open data no longer include the control member, so 2026 environment fields come from member 1.

Model changes. The ECMWF ensemble's raw intensity spread roughly doubled between 2024 and 2025, and HAFS replaced HWRF in 2023. The ensemble spread is set by the consensus error model rather than by the raw ensemble, so the first no longer affects it; aid availability changes are handled by leaving missing aids missing.

Statistical caveats. The two test seasons together contain about 60 storms, so confidence intervals are wide and only large differences are significant. The 2026 best tracks are NHC's working best tracks; post-season analysis often revises peak intensities. Hyperparameters were not tuned. The IFS-ENS gridded models have only three seasons of data. The day of the year is among the most important features in some models, which may partly reflect differences between seasons rather than physics.

Appendix F · Glossary and reproduction

CARQ NHC's operational analysis of position and intensity at t_0, with the preceding 24 h.

OFCL the official NHC forecast. HCCA: NHC's corrected consensus. IVCN: NHC's simple intensity consensus.

Early / late available / not available in time for the forecast cycle that starts at the synoptic time.

FNV3 (GDMN, GDMI) Google DeepMind's cyclone model; NHC files carry its ensemble mean (late) and an interpolated version (early).

CRPS continuous ranked probability score, the probabilistic analogue of the mean absolute error (lower is better).

TCCA, TCCB our consensus correction without and with SHIPS.

GECO GEFS Ensemble, Consensus-Oriented: NHC's GEFS member tracks around HCCA, full spread, every cycle.

IECO IFS-ENS Ensemble, Consensus-Oriented: our IFS-ENS members around HCCA, full spread, 12 UTC.

Reproduce. M1 checks: `scripts/m1_check.py`; consensus correction: `python -m tcic.consensus.features` and `scripts/m15_run.py`; gridded extraction: `scripts/pilot.sh`; gridded models: `python -m tcic.gridded.features` and `scripts/m2_run.py`; ensembles: `python -m tcic.ensemble.members`, `scripts/ens_run.py`, `scripts/ens_calib.py` (spread and RI models), `scripts/ens_smoothspread.py`, `scripts/ens_gridcentre.py` (gridded-centre variant), `scripts/ens_geco.py` (GECO and IECO), `scripts/`

`ens_ifs_hcca.py`, `scripts/ens_multiseason.py` (nine seasons), `scripts/fnv3_compare.py`; the whole chain: `scripts/rerun_chain.sh`; figures: `scripts/report_figures.py`, `scripts/report_figures_ens.py`, `scripts/case_plots.py`. Detailed notes are in `plan.md` and `docs/`.